Data Pipeline Reference

Technical reference for the Starburst Galaxy → Parquet → Ryugraph data pipeline.

Overview

This document details how relational data flows from **Starburst Galaxy** (managed Trino SaaS) SQL queries through Parquet files into Ryugraph/KuzuDB graph structures. It documents **external system behaviour** that our components depend on.

References:

- ADR-070: Starburst Galaxy + BigQuery Export Platform
- ADR-071: PyArrow Fallback Export Strategy

Related documents:

- requirements.md - Mapping definition JSON schema (authoritative source)
- export-worker.design.md - How our worker executes exports
- ryugraph-wrapper.design.md - How our wrapper loads graphs
- control-plane.design.md#mapping-generator-subsystem - How we validate mappings

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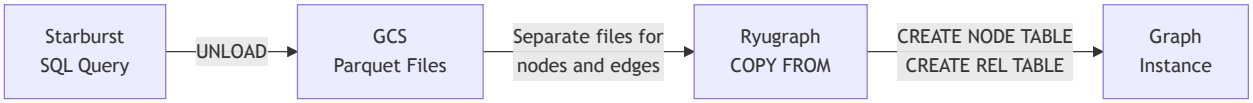
Data Flow Pipeline



▶ Mermaid Source

Two-Tier Export Strategy

The platform implements a two-tier export strategy with automatic fallback:

Tier	Method	When Used	Characteristics
1 (Primary)	Server-side (`system.unload()`)	Starburst Galaxy with GCS catalog	Distributed execution, direct GCS write
2 (Fallback)	Client-side (PyArrow)	When server-side unavailable	Streams through export worker, memory buffered

Export Flow with Fallback

```
Export Worker → Starburst Galaxy
                      │
          ┌───────────┴───────────┐
          │                       │
          ▼                       ▼
   system.unload()          SELECT * (fallback)
          │                       │
          │                       ▼
          │                Export Worker
          │                       │
          │                       ▼
          │                   PyArrow
          │                       │
          └──────────┬────────────┘
                     ▼
                   GCS
```

Fallback Triggers

The system falls back to PyArrow when:

- `system.unload()` is not available
- GCS catalog is not configured
- Feature flag is disabled
- Permission errors occur

Memory Considerations (Fallback Path)

Dataset Size	Memory Required	Recommendation
< 100 MB	256 MB	Safe for fallback
100 MB - 1 GB	1-2 GB	Monitor closely
> 1 GB	2+ GB	Prefer server-side

Starburst Galaxy UNLOAD

Syntax

The `UNLOAD` table function exports query results directly to GCS:

```
SELECT * FROM TABLE(
    io.unload(
        input => TABLE(
            SELECT customer_id, name, city, signup_date
            FROM analytics.customers
        ),
        location =>
'gs://bucket/{user_id}/{mapping_id}/{snapshot_id}/nodes/customers/',
        format => 'PARQUET',
        compression => 'SNAPPY'
    )
)
```

Parameters:

Parameter	Required	Description
input	Yes	TABLE(...) containing the SELECT query
location	Yes	GCS destination path
format	Yes	Output format (use 'PARQUET')
compression	No	Compression codec (recommend 'SNAPPY')

Parallel Writing Behaviour

Starburst writes in parallel **without** partitioning. Multiple worker nodes each produce their own output files:

```
gs://bucket/snapshot/nodes/customers/
├── 00000-123-uuid1-0-00001.parquet  ← worker 1
├── 00000-124-uuid2-0-00001.parquet  ← worker 2
├── 00000-125-uuid3-0-00001.parquet  ← worker 3
└── 00000-126-uuid4-0-00001.parquet  ← worker 4
```

File naming pattern: `{sequence}-{task}-{UUID}-{split}-{part}.parquet`

The number of files depends on cluster size and data volume. For typical ≤2GB graphs, this produces a manageable number of files.

Column Ordering Requirements

The Parquet schema is controlled entirely by the SELECT query. **Column ordering is critical:**

- For nodes: primary_key must be first column
- For edges: from_key, to_key must be first two columns
- Properties follow in defined order

```
SELECT * FROM TABLE(
    io.unload(
        input => TABLE(
            SELECT
                customer_id,                            -- Primary key (must
be first)
                name,                                   -- Property
                CAST(balance AS DOUBLE) as balance,     -- Cast DECIMAL to
DOUBLE
                signup_date                             -- Property
            FROM analytics.customers
        ),
        location => 'gs://bucket/path/nodes/customers/',
        format => 'PARQUET',
        compression => 'SNAPPY'
    )
)
```

Ryugraph Schema

Node Table Definition

```
CREATE NODE TABLE Customer(
    customer_id STRING PRIMARY KEY,
    name STRING,
    city STRING,
    signup_date DATE
);
```

Rules:

- Every node table requires a `PRIMARY KEY`
- Primary key must be unique across all nodes of that type
- Supported PK types: STRING, INT64, DATE, UUID

Relationship Table Definition

```
CREATE REL TABLE PURCHASED(
    FROM Customer TO Product,
    purchase_date DATE,
    amount DOUBLE,
    quantity INT64
);
```

Rules:

- Must specify `FROM NodeTable TO NodeTable`
- First two columns in Parquet must be FROM and TO primary keys
- Relationship properties follow the FROM/TO columns

Multi-Source/Target Relationships

```
CREATE REL TABLE REVIEWED(
    FROM Customer TO Product,
```

```
    FROM Customer TO Merchant,
    rating INT64,
    review_text STRING
);
```

When importing, specify which pair:

```
COPY REVIEWED FROM 'customer_product_reviews.parquet' (from='Customer',
to='Product');
COPY REVIEWED FROM 'customer_merchant_reviews.parquet' (from='Customer',
to='Merchant');
```

> **Design decision:** Our mapping schema does not currently support multi-source/target relationships. Each edge definition connects exactly one `from_node` to one `to_node`. For polymorphic relationships, define separate edge types (e.g., `REVIEWED_PRODUCT`, `REVIEWED_MERCHANT`). This simplifies validation, export, and import logic. Revisit if analysts need to query across polymorphic endpoints as a single relationship type—would require allowing multiple edge definitions with the same `type` and identical `properties`.

Multiplicity Constraints

```
-- Each customer has at most one primary address
CREATE REL TABLE HAS_PRIMARY_ADDRESS(FROM Customer TO Address, MANY_ONE);

-- One-to-one marriage relationship
CREATE REL TABLE MARRIED_TO(FROM Person TO Person, ONE_ONE);
```

> **Design decision:** Our mapping schema does not support multiplicity constraints. All relationships use the default `MANY_MANY`. Since our graphs are read-only analytics snapshots (no inserts/updates after initial load), cardinality enforcement provides no benefit—the source data already defines the actual cardinality.

Parquet File Requirements

Node Table Parquet Structure

For a node table:

```
CREATE NODE TABLE Customer(
    customer_id STRING PRIMARY KEY,
    name STRING,
    age INT64,
    city STRING
);
```

Parquet columns must be in exact order:

Column	Type	Notes
customer_id	STRING	Primary key (required, first)
name	STRING	Property
age	INT64	Property
city	STRING	Property

Relationship Table Parquet Structure

For a relationship table:

```
CREATE REL TABLE PURCHASED(
    FROM Customer TO Product,
    purchase_date DATE,
    amount DOUBLE
);
```

Parquet columns must be:

Column	Type	Notes
customer_id	STRING	FROM node primary key (required, first)
product_id	STRING	TO node primary key (required, second)

Column	Type	Notes
purchase_date	DATE	Relationship property
amount	DOUBLE	Relationship property

SERIAL Primary Keys (Auto-Increment)

For large datasets with sequential IDs:

```
CREATE NODE TABLE Event(
    id SERIAL PRIMARY KEY,
    event_type STRING,
    timestamp TIMESTAMP
);
```

When using SERIAL:

- **Omit the primary key column from Parquet file**
- Ryugraph auto-generates IDs starting from 0
- Significantly improves load performance

Parquet columns for SERIAL node:

Column	Type
event_type	STRING
timestamp	TIMESTAMP

> **Design decision:** Our mapping schema does not support SERIAL primary keys. We always use the source database primary key to maintain traceability between graph nodes and source data. At our target scale (≤2GB graphs, ~10M nodes max), the performance benefit of SERIAL is negligible. SERIAL would also complicate edge definitions since edges reference nodes by primary key, and auto-generated IDs aren't known until after node import. Revisit if scale increases significantly (50M+ nodes)—SERIAL provides fastest graph traversal via sequential INT64 keys with optimal memory layout and cache locality.

Ryugraph COPY FROM

Syntax

```
COPY TableName FROM 'gs://bucket/path/*.parquet';
```

Import Order Constraint

Critical: Always import nodes before relationships. Relationships reference node primary keys that must already exist.

```
-- STEP 1: Import nodes FIRST (order matters!)
COPY Customer FROM 'gs://bucket/snapshot_123/nodes/customers/*.parquet';
COPY Product FROM 'gs://bucket/snapshot_123/nodes/products/*.parquet';

-- STEP 2: Import relationships AFTER nodes exist
COPY PURCHASED FROM 'gs://bucket/snapshot_123/edges/purchased/*.parquet';
```

Glob Pattern Support

Ryugraph reads all matching files via glob pattern:

```
COPY Customer FROM 'gs://bucket/path/*.parquet';  -- reads all parquet files
```

Type Mapping

Starburst to Ryugraph Type Mapping

Starburst/Trino Type	Ryugraph Type	Notes
VARCHAR, CHAR	STRING	UTF-8 encoded

Starburst/Trino Type	Ryugraph Type	Notes
BIGINT	INT64	64-bit signed integer
INTEGER	INT32	32-bit signed integer
SMALLINT	INT16	16-bit signed integer
TINYINT	INT8	8-bit signed integer
DOUBLE	DOUBLE	64-bit floating point
REAL	FLOAT	32-bit floating point
DECIMAL	DOUBLE	Cast in SQL query
DATE	DATE	Calendar date
TIMESTAMP	TIMESTAMP	Date and time
TIMESTAMP WITH TIME ZONE	TIMESTAMP	Timezone stripped
BOOLEAN	BOOL	True/false
VARBINARY	BLOB	Binary data
ARRAY	LIST	Variable-length array
MAP	MAP	Key-value pairs
ROW	STRUCT	Structured type
UUID	UUID	Universal identifier

Required SQL Casts

Some types require explicit casting in the SQL query:

```
SELECT
    customer_id,
    name,
    CAST(price AS DOUBLE) as price,           -- DECIMAL -> DOUBLE
    CAST(created_at AS TIMESTAMP) as created, -- Handle timezone
```

```
    CAST(is_active AS BOOLEAN) as is_active   -- Ensure boolean
FROM source_table
```

Error Handling

Duplicate Primary Keys

Ryugraph rejects duplicate primary keys by default:

```
-- Enable error skipping for dirty data
COPY Customer FROM 'file.parquet' (IGNORE_ERRORS=true);
```

Note: `IGNORE_ERRORS=true` has performance cost. Prefer clean data.

Missing Foreign Keys

Relationships referencing non-existent nodes will fail:

```
Error: Node with primary key 'customer_xyz' not found in Customer table
```

Solution: Always load nodes before relationships.

Schema Mismatch

Parquet columns must match Ryugraph schema exactly:

```
Error: Column count mismatch. Expected 4 columns, got 3.
```

Solution: Ensure SELECT column order matches CREATE TABLE property order.

Performance Considerations

Parallel I/O

Stage	Parallelism	How
Starburst UNLOAD	Multiple workers write files concurrently	Automatic based on cluster size
Ryugraph COPY FROM	Multiple threads read files concurrently	`max_num_threads` setting

Both parallel write and parallel read work without partitioning. The flat file structure is optimal:

```
gs://bucket/snapshot/nodes/customers/
├── file-001.parquet  ← written by worker 1, read by thread A
├── file-002.parquet  ← written by worker 2, read by thread B
├── file-003.parquet  ← written by worker 3, read by thread C
└── ...
```

Starburst Export Tuning

- Use `compression => 'SNAPPY'` for balance of speed and size
- Files are automatically parallelised across workers
- No partitioning needed for performance

Ryugraph Import Tuning

- **COPY FROM is fastest** for bulk loading (vs individual inserts)
- Use glob patterns: `COPY FROM 'path/*.parquet'` to read all files
- Ryugraph reads multiple files in parallel using available threads
- Set buffer pool appropriately: `buffer_pool_size=2_147_483_648` (2GB)
- For consecutive integer IDs, use `SERIAL` and omit PK column

Memory Management

For large graphs (approaching 2GB limit):

- Configure Ryugraph buffer pool to ~80% of pod memory
- Enable disk spilling for larger-than-memory operations
- Use persistent volume for spill files

Mapping Many-to-Many Relationships

In relational databases, many-to-many relationships are represented using **join tables** with foreign keys to both entities. In the graph model, these become **relationship tables with properties**.

Relational Pattern

```sql
-- Join table with foreign keys to both entities
CREATE TABLE transactions (
    transaction_id VARCHAR PRIMARY KEY,
    customer_id VARCHAR REFERENCES customers(customer_id),
    product_id VARCHAR REFERENCES products(product_id),
    amount DECIMAL(10,2),
    transaction_date TIMESTAMP
);
```

Graph Pattern

```
-- Foreign keys become FROM/TO, other columns become properties
CREATE REL TABLE PURCHASED(
    FROM Customer TO Product,
    transaction_id STRING,
    amount DOUBLE,
    transaction_date TIMESTAMP
);
```

Mapping Rules

Relational Concept	Graph Equivalent
Join table	REL TABLE
FK to source entity	FROM NodeType
FK to target entity	TO NodeType
Other columns	Relationship properties
Join table PK	Optional property (or omit if not needed)

Export Query Structure

Critical: Column ordering determines the relationship direction.

```
SELECT
    customer_id,           -- FROM node PK (must be first)
    product_id,            -- TO node PK (must be second)
    transaction_id,        -- Property
    CAST(amount AS DOUBLE) as amount,
    transaction_date
FROM transactions
```

Import Order

Always load **both node tables before** the relationship table:

```
-- 1. Load nodes first
COPY Customer FROM 'gs://bucket/nodes/Customer/*.parquet';
COPY Product FROM 'gs://bucket/nodes/Product/*.parquet';

-- 2. Load relationships after nodes exist
COPY PURCHASED FROM 'gs://bucket/edges/PURCHASED/*.parquet';
```

Complete Example

Source: Relational Schema (in Starburst)

```sql
-- Customers table
CREATE TABLE customers (
    customer_id VARCHAR PRIMARY KEY,
    name VARCHAR,
    email VARCHAR,
    city VARCHAR,
    signup_date DATE
);

-- Products table
CREATE TABLE products (
    product_id VARCHAR PRIMARY KEY,
    name VARCHAR,
    category VARCHAR,
    price DECIMAL(10,2)
);

-- Transactions table (join table with attributes)
CREATE TABLE transactions (
    transaction_id VARCHAR PRIMARY KEY,
    customer_id VARCHAR REFERENCES customers(customer_id),
    product_id VARCHAR REFERENCES products(product_id),
    amount DECIMAL(10,2),
    transaction_date TIMESTAMP
);
```

Target: Graph Schema (in Ryugraph)

```sql
-- Node tables
CREATE NODE TABLE Customer(
    customer_id STRING PRIMARY KEY,
    name STRING,
    email STRING,
    city STRING,
    signup_date DATE
);

CREATE NODE TABLE Product(
    product_id STRING PRIMARY KEY,
    name STRING,
    category STRING,
    price DOUBLE
);

-- Relationship table (foreign keys become explicit edges)
CREATE REL TABLE PURCHASED(
    FROM Customer TO Product,
```

```
    transaction_id STRING,
    amount DOUBLE,
    transaction_date TIMESTAMP
);
```

Export Queries

```sql
-- Export Customers
SELECT * FROM TABLE(io.unload(
    input => TABLE(
        SELECT customer_id, name, email, city, signup_date
        FROM analytics.customers
    ),
    location => 'gs://bucket/snapshot_123/nodes/Customer/',
    format => 'PARQUET',
    compression => 'SNAPPY'
))

-- Export Products
SELECT * FROM TABLE(io.unload(
    input => TABLE(
        SELECT product_id, name, category, CAST(price AS DOUBLE) as price
        FROM analytics.products
    ),
    location => 'gs://bucket/snapshot_123/nodes/Product/',
    format => 'PARQUET',
    compression => 'SNAPPY'
))

-- Export PURCHASED relationships
SELECT * FROM TABLE(io.unload(
    input => TABLE(
        SELECT
            t.customer_id,           -- FROM node PK (first)
            t.product_id,            -- TO node PK (second)
            t.transaction_id,        -- Property
            CAST(t.amount AS DOUBLE) as amount,
            t.transaction_date
        FROM analytics.transactions t
    ),
    location => 'gs://bucket/snapshot_123/edges/PURCHASED/',
    format => 'PARQUET',
    compression => 'SNAPPY'
))
```

Import Commands

```
-- Create schema
CREATE NODE TABLE Customer(customer_id STRING PRIMARY KEY, name STRING, email
STRING, city STRING, signup_date DATE);
CREATE NODE TABLE Product(product_id STRING PRIMARY KEY, name STRING, category
STRING, price DOUBLE);
CREATE REL TABLE PURCHASED(FROM Customer TO Product, transaction_id STRING,
amount DOUBLE, transaction_date TIMESTAMP);

-- Import nodes first
COPY Customer FROM 'gs://bucket/snapshot_123/nodes/Customer/*.parquet';
COPY Product FROM 'gs://bucket/snapshot_123/nodes/Product/*.parquet';

-- Import relationships after nodes exist
COPY PURCHASED FROM 'gs://bucket/snapshot_123/edges/PURCHASED/*.parquet';
```

GCS File Structure

```
gs://bucket/snapshot_123/
├── nodes/
│   ├── Customer/
│   │   ├── 00000-xxx-uuid1.parquet
│   │   └── 00000-xxx-uuid2.parquet
│   └── Product/
│       └── 00000-xxx-uuid3.parquet
└── edges/
    └── PURCHASED/
        ├── 00000-xxx-uuid4.parquet
        └── 00000-xxx-uuid5.parquet
```

Starburst/Trino REST API Protocol

This section documents the Starburst REST API used for async query execution.

Reference: Trino Client Protocol.

Query Submission

Submit a query via POST to `/v1/statement`:

```http
POST /v1/statement HTTP/1.1
Host: starburst.example.com
X-Trino-Catalog: analytics
X-Trino-Schema: public
Content-Type: text/plain
Authorization: Basic <base64>

SELECT * FROM TABLE(io.unload(...))
```

Response (success):

```json
{
  "id": "20250116_123456_00001_abcde",
  "infoUri": "http://starburst:8080/ui/query/20250116_123456_00001_abcde",
  "nextUri": "http://starburst:8080/v1/query/20250116_123456_00001_abcde/1",
  "stats": {
    "state": "QUEUED",
    "queued": true,
    "scheduled": false,
    "nodes": 0,
    "totalSplits": 0,
    "queuedSplits": 0,
    "runningSplits": 0,
    "completedSplits": 0
  }
}
```

Response (immediate error):

```json
{
  "id": "20250116_123456_00001_abcde",
  "stats": {"state": "FAILED"},
  "error": {
    "message": "line 1:1: Table 'analytics.public.invalid' does not exist",
    "errorCode": 1,
    "errorName": "TABLE_NOT_FOUND",
    "errorType": "USER_ERROR"
  }
}
```

Query Polling

Poll query status via GET to the `nextUri`:

```http
GET /v1/query/20250116_123456_00001_abcde/1 HTTP/1.1
Host: starburst.example.com
Authorization: Basic <base64>
```

Response (in progress):

```json
{
  "id": "20250116_123456_00001_abcde",
  "nextUri": "http://starburst:8080/v1/query/20250116_123456_00001_abcde/2",
  "stats": {
    "state": "RUNNING",
    "queued": false,
    "scheduled": true,
    "nodes": 4,
    "totalSplits": 100,
    "queuedSplits": 20,
    "runningSplits": 30,
    "completedSplits": 50
  }
}
```

Response (finished - no nextUri):

```json
{
  "id": "20250116_123456_00001_abcde",
  "stats": {
    "state": "FINISHED",
    "queued": false,
    "scheduled": true,
    "nodes": 4,
    "totalSplits": 100,
    "queuedSplits": 0,
    "runningSplits": 0,
    "completedSplits": 100
  }
}
```

Response (failed):

```json
{
  "id": "20250116_123456_00001_abcde",
  "stats": {"state": "FAILED"},
```

```
  "error": {
    "message": "Query exceeded maximum time limit of 1.00h",
    "errorCode": 65540,
    "errorName": "EXCEEDED_TIME_LIMIT",
    "errorType": "USER_ERROR"
  }
}
```

Query State Machine

```
     QUEUED
        ↓
    PLANNING
        ↓
    STARTING
        ↓
     RUNNING  ———→  FAILED
        ↓
   FINISHING
        ↓
    FINISHED
```

State descriptions:

State	Description
`QUEUED`	Query waiting in queue
`PLANNING`	Query plan being generated
`STARTING`	Allocating resources
`RUNNING`	Executing on workers
`FINISHING`	Completing final operations
`FINISHED`	Query completed successfully
`FAILED`	Query failed (check `error` field)

Protocol Rules

1. **Completion detection**: Keep polling `nextUri` until it's **absent** from the response. The `state` field is for humans only.
2. **HTTP error handling**:

Status	Action
200	Process response
429	Retry after `Retry-After` header
502, 503, 504	Retry in 50-100ms (load balancer issue)
Other	Query failed

3. **Authentication**: Headers only required on initial POST, not when following `nextUri`.
4. **Query cancellation**: Send `DELETE` to `nextUri` to cancel a running query.

Required Headers

Header	Required	Description
`X-Trino-Catalog`	Yes (POST)	Default catalog
`X-Trino-Schema`	Yes (POST)	Default schema
`Authorization`	Yes	Basic auth or Bearer token
`Content-Type`	Yes (POST)	`text/plain` for query body

Appendix: GCS Configuration for Starburst

Catalog Configuration

```
connector.name=hive
hive.metastore.uri=thrift://metastore:9083
```

```
# Enable native GCS
fs.native-gcs.enabled=true
gcs.project-id=your-gcp-project
gcs.json-key-file-path=/path/to/service-account.json
```

UNLOAD Credentials

```
{
  "id": "gcs-export",
  "location": "gs://your-bucket/",
  "configuration": {
    "fs.native-gcs.enabled": "true",
    "gcs.json-key": "{...service account key JSON...}"
  }
}
```

Required GCS Permissions

- `storage.objects.create`
- `storage.objects.delete` (for overwrites)
- `storage.objects.get`
- `storage.objects.list`
- `storage.buckets.get`

E2E Testing with Trino

For E2E testing, we use open-source Trino with a translation proxy instead of Starburst Enterprise. This approach replicates production behavior by executing real SQL queries and writing actual Parquet files.

Architecture

Starburst's `io.unload()` is **proprietary** and not available in open-source Trino. The trino-proxy translates `io.unload()` calls to Hive CTAS (CREATE TABLE AS SELECT) with `external_location`:

```sql
-- Input (Starburst io.unload):
SELECT * FROM TABLE(io.unload(
    input => TABLE(SELECT id, name FROM catalog.schema.table),
    location => 'gs://bucket/export/',
    format => 'PARQUET', compression => 'SNAPPY'))

-- Output (Trino Hive CTAS):
CREATE TABLE hive.temp.export_abc123
WITH (format = 'PARQUET', external_location = 'gs://bucket/export/')
AS SELECT id, name FROM catalog.schema.table
```

Components

Component	Image	Purpose
trino	`trino-gcs:e2e-test`	Trino 479 with native GCS (`gcs.endpoint` for fake-gcs-server)
trino-proxy	`trino-proxy:e2e-test`	io.unload() → Hive CTAS translation
hive-metastore	`hive-metastore-gcs:e2e-test`	Apache Hive 4.1.0 with GCS connector JAR
fake-gcs-server	`fsouza/fake-gcs-server:1.52.3`	GCS emulator for local testing

Why This Approach

1. **Same execution path** - Trino executes queries and writes Parquet, same as production
2. **Same storage API** - Uses GCS URLs (`gs://`), not S3/MinIO
3. **Same file format** - Raw Parquet files, not Iceberg tables
4. **Export-worker unchanged** - No code changes needed for E2E testing

References

- ADR-029: E2E Trino Stack with Hive CTAS Translation

- e2e-tests.design.md
- tests/e2e/k8s/ - E2E test Kubernetes configurations